Week 7: Effect Modification

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R Packages

```
# Load required packages using pacman
pacman::p_load(
  dagitty, # DAG construction and analysis
  ggdag, # Beautiful DAG plotting with ggplot2
  tidyverse, # Data manipulation and visualization
  broom, # Tidy model outputs
  grf, # Generalized random forests for causal ML
  rpart, # Decision trees for causal trees
  rpart.plot, # Plotting decision trees
  kableExtra, # Enhanced tables
  gt, # Grammar of tables
  patchwork, # Combining plots
  marginaleffects # Marginal effects and contrasts
)
```

Learning Objectives

By the end of this week, students will be able to:

1. **Define** effect modification and distinguish it from confounding using formal notation and potential outcomes framework

2. **Compare** additive versus multiplicative scales of interaction and explain when each is most appropriate
3. **Differentiate** between statistical interaction and mechanistic interaction using Rothman's sufficient cause framework
4. **Calculate** measures of effect modification by hand using simple 2×2 tables and potential outcomes
5. **Implement** R code to detect, visualize, and quantify effect modification in observational data
6. **Apply** modern causal machine learning methods (causal trees, causal forests) to discover heterogeneous treatment effects
7. **Interpret** interaction terms in regression models within a causal framework

1 Comprehensive Topic Explanation

1.1 What is Effect Modification?

Effect modification, also known as **treatment effect heterogeneity**, occurs when the causal effect of an exposure varies across different levels or values of another variable, called the **effect modifier** (Hernán & Robins, 2025, Chapter 4). Unlike confounding, which creates bias that must be controlled, effect modification represents genuine biological or social phenomena where treatment effects differ across subgroups.

Formally, let V be a potential effect modifier. We say that V modifies the effect of treatment A on outcome Y if:

$$\mathbb{E}[Y^{a=1} - Y^{a=0} \mid V = v] \neq \mathbb{E}[Y^{a=1} - Y^{a=0} \mid V = v']$$

for some values $v \neq v'$ of the modifier V.

This definition is **inherently causal** because it compares potential outcomes within strata of the modifier. Effect modification cannot be definitively established from purely observational associations—it requires the same causal assumptions (exchangeability, positivity, consistency) that we need for identifying any causal effect.

1.2 Scales of Interaction: Additive vs. Multiplicative

The same data can show interaction on one scale but not another, leading to the crucial question: **which scale matters?**

Additive Scale: Interaction exists if the joint effect differs from the sum of individual effects:

$$\mathbb{E}[Y^{a=1,v=1}]-\mathbb{E}[Y^{a=0,v=0}] \neq (\mathbb{E}[Y^{a=1,v=0}]-\mathbb{E}[Y^{a=0,v=0}])+(\mathbb{E}[Y^{a=0,v=1}]-\mathbb{E}[Y^{a=0,v=0}])$$

Multiplicative Scale: Interaction exists if the joint effect differs from the product of individual relative effects:

$$\frac{\mathbb{E}[Y^{a=1,v=1}]}{\mathbb{E}[Y^{a=0,v=0}]} \neq \frac{\mathbb{E}[Y^{a=1,v=0}]}{\mathbb{E}[Y^{a=0,v=0}]} \times \frac{\mathbb{E}[Y^{a=0,v=1}]}{\mathbb{E}[Y^{a=0,v=0}]}$$

For **rare outcomes**, the additive scale is often more relevant for public health because it directly quantifies excess cases attributable to interaction. For **common outcomes** or when thinking about biological mechanisms, multiplicative interactions may be more natural.

1.3 Statistical vs. Mechanistic Interaction

Statistical interaction refers to departure from additivity (or multiplicativity) in a fitted model. It's what we observe in regression output when interaction terms are significant.

Mechanistic interaction refers to biological or causal pathways where factors truly work together to produce effects. Rothman's sufficient cause framework provides the clearest illustration of mechanistic interaction.

1.4 Rothman's Sufficient Cause Framework

Kenneth Rothman introduced "causal pies" to represent sufficient causes—combinations of factors that, together, inevitably produce the outcome (VanderWeele, 2015, Chapter 1). Each "pie" represents one pathway to disease, with each "slice" representing a component cause.

Synergism occurs when two factors appear as component causes in the same causal pie. For example:

- Pie 1: {Smoking, Asbestos exposure, Genetic susceptibility} → Lung cancer
- Pie 2: {Heavy alcohol use, Hepatitis B, Poor nutrition} → Liver disease

If smoking and asbestos appear together in Pie 1 but neither appears in other pies, they exhibit synergistic interaction.

This framework explains why:

- Interaction can exist on the additive scale even when multiplicative interaction is absent
- Mechanistic interaction may not always manifest as statistical interaction
- The "background" prevalence of other component causes affects observed interaction patterns

1.5 Causal Interaction with Multiple Treatments

When we have two treatments A and B, we can define four potential outcomes for each individual:

- $Y^{a=0,b=0}$: Neither treatment
- $Y^{a=1,b=0}$: Only treatment A
- $Y^{a=0,b=1}$: Only treatment B
- $Y^{a=1,b=1}$: Both treatments

Additive causal interaction exists when:

$$\mathbb{E}[Y^{a=1,b=1} - Y^{a=0,b=0}] \neq (\mathbb{E}[Y^{a=1,b=0} - Y^{a=0,b=0}] + \mathbb{E}[Y^{a=0,b=1} - Y^{a=0,b=0}])$$

The left side represents the **joint effect**, while the right side represents the **sum of marginal effects**.

1.6 Modern Causal Machine Learning Approaches

Traditional approaches to effect modification require **a priori** specification of interaction terms. Modern causal machine learning methods can **discover** heterogeneous treatment effects without pre-specifying interactions.

Causal Trees: Recursive partitioning that splits on covariates to maximize treatment effect heterogeneity rather than outcome prediction (Athey & Imbens, 2016).

Causal Forests: An ensemble method that averages many causal trees, providing both point estimates and uncertainty quantification for heterogeneous treatment effects (Wager & Athey, 2018).

These methods are particularly valuable when:

- The modifier space is high-dimensional
- Interactions involve multiple variables simultaneously
- The functional form of modification is unknown

2 Intuitive Clarifications of Challenging Ideas

□ Learning Tip: The Interaction Paradox

Analogy: Think of effect modification like cooking. Salt enhances the flavor of tomatoes more than it enhances the flavor of ice cream. The "effect" of salt depends on what you're adding it to—this is effect modification.

However, whether we measure this on a "taste intensity" scale (additive) or a "flavor multiplication" scale (multiplicative) can lead to different conclusions about whether interaction exists.

2.1 Why Confounding Effect Modification

Common Misconception: "If a variable is associated with both treatment and outcome, it's a confounder."

Reality: Effect modifiers can be associated with treatment and outcome WITHOUT being confounders. The key difference:

- **Confounder**: Creates bias in the marginal treatment effect
- **Effect Modifier**: Changes the treatment effect across strata, but each stratum-specific effect may be unbiased

Visual Metaphor: Think of confounding as wearing tinted glasses that distort ALL colors equally. Effect modification is like having different lighting in different rooms—the same object looks different in each room, but you're seeing it clearly in each context.

2.2 The Scale Dependence Problem

Consider these data:

Group	Risk with Treatment	Risk without Treatment	Risk Difference	Risk Ratio
Men	0.20	0.10	0.10	2.0
Women	0.15	0.05	0.10	3.0

- **Additive scale**: No interaction (both differences = 0.10)
- **Multiplicative scale**: Strong interaction (ratios = 2.0 vs 3.0)

Intuition: This happens because the baseline risks differ between men and women. When baseline risk is lower, the same absolute increase represents a larger relative increase.

2.3 Why Sufficient Causes Matter

Mechanical Analogy: Think of disease as a car failing to start. There might be multiple ways for the car to fail:

- Path 1: {Dead battery} + {Cold weather} + {Old alternator}

- Path 2: {Empty gas tank} + {Clogged fuel filter}
- Path 3: {Faulty starter} + {Corroded connections}

If "cold weather" and "dead battery" appear together in Path 1, they're synergistic—cold weather only matters when you also have a dead battery problem.

This explains why some interactions only appear in certain populations or contexts.

3 Simple, Hand-Calculation Numerical Example

Let's work through a complete example calculating effect modification by hand.

Worked Example: Drug Efficacy by Age Group

Scenario: A new antihypertensive drug is tested. We want to know if age modifies its effect.

Data:

Age Group	Treatment	n	Mean BP Reduction
Young (<50)	Drug	100	15 mmHg
Young (<50)	Placebo	100	3 mmHg
Old (≥50)	Drug	100	20 mmHg
Old (≥50)	Placebo	100	8 mmHg

Step 1: Calculate stratum-specific effects

Young group effect: $15 - 3 = 12$ mmHg reduction

Old group effect: $20 - 8 = 12$ mmHg reduction

Step 2: Test for effect modification

Since both effects equal 12 mmHg, there is **no effect modification on the additive scale**.

Step 3: Check multiplicative scale

Young group ratio: $15/3 = 5.0$

Old group ratio: $20/8 = 2.5$

Since $5.0 \neq 2.5$, there **is effect modification on the multiplicative scale**.

Interpretation: The drug provides the same absolute benefit (12 mmHg) regardless of age, but represents a larger relative improvement in younger patients due to their lower baseline response to placebo.

Hand Calculation: Sufficient Cause Framework

Scenario: Consider lung cancer with two exposures: smoking (S) and asbestos (A).

Assume three sufficient cause "pies":

- Pie 1: {S, A, U□} (smoking + asbestos + unknown factors)
- Pie 2: {S, U□, U□} (smoking + other unknown factors)
- Pie 3: {U□, U□, U□} (non-smoking causes)

Population prevalences:

- S: 30%, A: 10%, U□: 5%, U□: 2%, U□: 40%, U□: 1%, U□: 20%, U□: 30%

Step 1: Calculate disease probabilities

$$
\begin{aligned}
\mathbb{P}(\text{Disease}|\text{Disease}|S=1,A=1) &= \mathbb{P}(\text{Pie 1} \vee \text{Pie 2}) \\
&= \mathbb{P}(\text{Pie 1}) + \mathbb{P}(\text{Pie 2}) - \mathbb{P}(\text{Pie 1} \wedge \text{Pie 2}) \\
&= \mathbb{P}(U_1 = 1) + \mathbb{P}(U_2 = 1 \wedge U_3 = 1) - \mathbb{P}(U_1 = 1 \wedge U_2 = 1 \wedge U_3 = 1) \\
&= 0.05 + (0.02 \times 0.40) - (0.05 \times 0.02 \times 0.40) \\
&= 0.05 + 0.008 - 0.0004 \\
&= 0.0576
\end{aligned} \tag{1}
$$

$$\mathbb{P}(\text{Disease}|S=1,A=0) = \mathbb{P}(\text{Pie 2}) = \mathbb{P}(U_2 = 1 \wedge U_3 = 1) = 0.02{\times}0.40 = 0.008$$

$$\mathbb{P}(\text{Disease}|S=0,A=1) = \mathbb{P}(\text{Pie 3}) = \mathbb{P}(U_4 = 1 \wedge U_5 = 1 \wedge U_6 = 1) = 0.01{\times}0.20{\times}0.30 = 0.0006$$

$$\mathbb{P}(\text{Disease}|S=0,A=0) = \mathbb{P}(\text{Pie 3}) = 0.0006$$

Step 2: Calculate interaction measures

Joint effect: 0.0576 - 0.0006 = 0.057

Smoking alone: 0.008 - 0.0006 = 0.0074

Asbestos alone: 0.0006 - 0.0006 = 0

Sum of individual effects: 0.0074 + 0 = 0.0074

Synergy index: (0.057 - 0.0074)/0.0074 = 6.7

This indicates strong synergistic interaction—the combined effect is nearly 7 times larger than expected from adding individual effects.

4 R Code Examples & Interpretation

4.1 Simulating Effect Modification Data

```r
# Set seed for reproducibility
set.seed(12345)

# Simulate data with age-dependent treatment effect
n <- 1000
age <- rnorm(n, mean = 50, sd = 15)
age_centered <- age - 50

# Treatment assignment (slightly more likely for older patients)
treatment <- rbinom(n, 1, prob = 0.4 + 0.001 * age_centered)

# Outcome with effect modification
# Base outcome depends on age
# Treatment effect is stronger in older patients
baseline_outcome <- 120 + 0.5 * age_centered + rnorm(n, 0, 10)
treatment_effect <- -10 - 0.2 * age_centered # Stronger effect in older patients
outcome <- baseline_outcome + treatment * treatment_effect

# Create dataset
sim_data <- tibble(
  id = 1:n,
  age = age,
  age_group = ifelse(age >= 50, "Older", "Younger"),
  treatment = factor(treatment, labels = c("Control", "Treatment")),
  outcome = outcome
)
```

```r
# Show first few rows
head(sim_data) ▷
  kable(digits = 2, caption = "Simulated Dataset with
  Effect Modification") ▷
  kable_styling(bootstrap_options = "striped")
```

Table 3: Simulated Dataset with Effect Modification

id	age	age_group	treatment	outcome
1	58.78	Older	Treatment	110.89
2	60.64	Older	Treatment	106.49
3	48.36	Younger	Control	124.25
4	43.20	Younger	Treatment	120.43
5	59.09	Older	Control	112.06
6	22.73	Younger	Treatment	82.47

4.2 Exploratory Analysis of Effect Modification

```r
# Visualize potential effect modification
p1 <- sim_data ▷
  ggplot(aes(x = treatment, y = outcome, fill =
  treatment)) +
  geom_boxplot(alpha = 0.7) +
  facet_wrap(~age_group) +
  labs(
    title = "Treatment Effects by Age Group",
    y = "Blood Pressure (mmHg)",
    x = "Treatment Group"
  ) +
  theme_minimal() +
  scale_fill_brewer(type = "qual", palette = "Set2") +
  theme(legend.position = "none")
```

```r
# Scatter plot with regression lines
p2 <- sim_data  ▷
  ggplot(aes(x = age, y = outcome, color = treatment)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Treatment Effect Varies with Age",
    x = "Age (years)",
    y = "Blood Pressure (mmHg)",
    color = "Treatment"
  ) +
  theme_minimal() +
  scale_color_brewer(type = "qual", palette = "Set1")

# Combine plots
p1 / p2
```

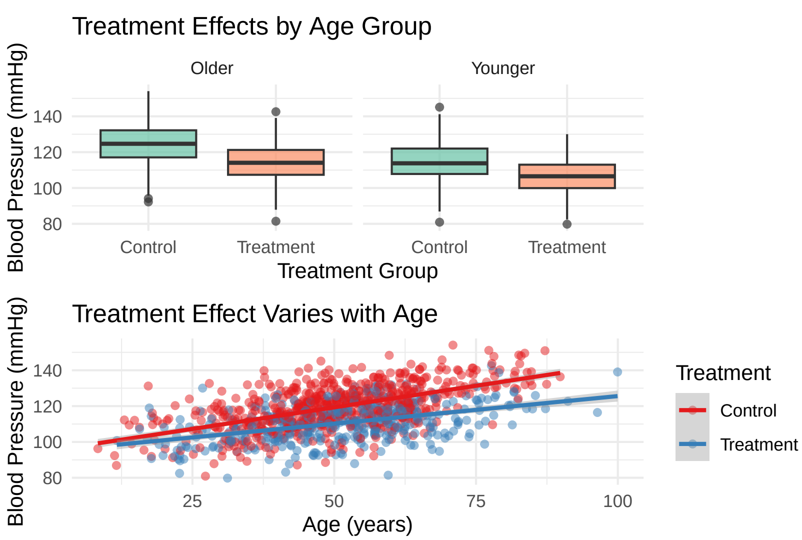

Figure 1: Exploring Effect Modification Patterns

This visualization immediately suggests that the treatment effect differs between age groups. The older group shows a larger treatment effect (greater separation between treatment and control).